

Prompts Are Design Instructions

Project: Helpful Classroom Sorter

1. Your mission

Your learning target

I can write a prompt with a clear task, supplied evidence and checkable output requirements.

I can test a response against criteria chosen before seeing it.

I can revise a prompt using observed failures while preserving missing and conflicting information.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

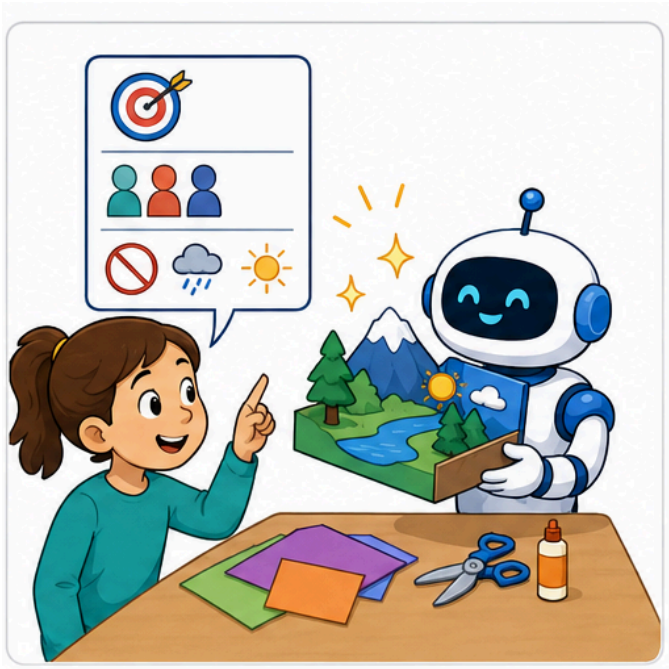
Without opening the old lesson, explain one key idea from Generate, Retrieve, or Calculate?. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Recommendations. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Clear goals, context, and constraints improve usefulness, not truth guarantees.



LEARNING TARGET Write a request for a three-bullet explanation of evaporation for Grade 6 using only a supplied paragraph. Create three separately checkable acceptance criteria. What should happen if the paragraph does not explain evaporation?

Learner name:

Readiness for the connected explanation

A generated invitation adds a meeting time that is absent from its source. Does fluent wording make that time supported? What should happen next?

This week's end goal

By the end, I can write a prompt contract that handles absent and conflicting facts.

2. Notice the evidence

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; evidence tells us what the clues mean.



MYSTERY 'Write a story' gives little direction; a goal, audience, setting, length, required facts, and safety constraints shape a more useful output.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first prediction and state the clue that supports it.

My observations, prediction, and evidence:

GOOD THINKING A prediction can change when stronger evidence appears.

A closer explanation

"Write something good about our club" leaves many decisions unstated. Who will read it? Which facts may be used? What must the output contain? A more useful prompt states the job, audience, evidence and requirements. This makes it easier to judge the result rather than choosing the answer that sounds most confident.

Use this fictional source C: the paper modelling club meets Monday in Room 7, and participants bring paper. C gives no start time. Our task is a short information card for Grade 6 readers. It must report day, time, place and what to bring, using only C. There is no need to supply names or private information.

Prompt P1: Using only source C, produce an information card for Grade 6 readers. Return exactly four labelled lines in this order: Day, Time, Place, Bring. Copy or faithfully restate each supported value. For a

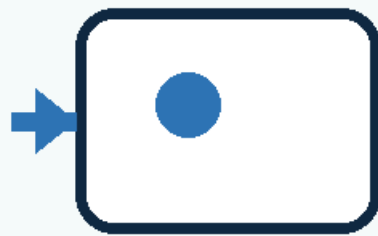
missing value write “not stated”. Add no other details. Source C: the paper modelling club meets Monday in Room 7; bring paper.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



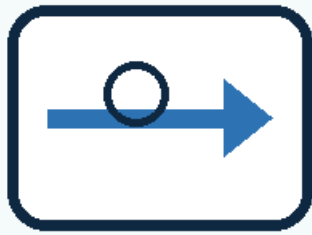
INPUT

goal, context, constraints, and requested
format



PROCESS

a generative model predicts an output
matching the instruction



OUTPUT

draft that still requires review

INPUT goal, context, constraints, and requested format

PROCESS a generative model predicts an output matching the instruction

OUTPUT draft that still requires review

CORE IDEA Clear goals, context, and constraints improve usefulness, not truth guarantees.

UNDER THE HOOD Prompting conditions model behavior but does not change underlying knowledge, eliminate hallucination, or grant access to missing sources. Strong prompts specify success criteria and evidence boundaries, then support iterative evaluation.

Draw a fresh example with all three stages:

A closer explanation

From prompt requirement to evidence of success

Requirement · Expected behaviour · How to check

Requirement: Use source C only

Expected behaviour: Monday and Room 7

How to check: Compare with C

Requirement: Mark missing values

Expected behaviour: Time: not stated

How to check: Check absent time

Requirement: Four labelled lines

Expected behaviour: Day Time Place Bring

How to check: Count and inspect

Requirement: No added details

Expected behaviour: No invented promises

How to check: Check every claim

Choose these checks before reading a response. A response must satisfy all required criteria, not merely look tidy.

For P1, an acceptable card has Day: Monday; Time: not stated; Place: Room 7; Bring: paper, each on its own line. The semicolons here separate lines for discussion. A correct format with an invented 3 pm time still fails the evidence requirement. Accurate facts inside a long paragraph still fail the requested four-line format.

These are classroom response examples written for inspection, not measurements from a live AI system. You can test the instructions by asking a partner to follow them on paper. That checks how clear the instructions are, but it does not prove that a model will always follow them.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

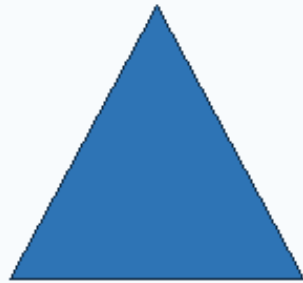
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the evidence carefully.



Write a story



**300-word space mystery
for age 8**



Use only these facts



Be accurate

CASE 1 — Write a story Clue: goal underspecified. Meaning: vague prompt.

CASE 2 — 300-word space mystery for age 8 Clue: audience and format clear. Meaning: better direction.

CASE 3 — Use only these facts Clue: evidence boundary. Meaning: still verify.

CASE 4 — Be accurate Clue: wish without checking plan. Meaning: not a guarantee.

BOUNDARY Prompt quality improves control and usefulness, not guaranteed truth.

Which two pictures are easiest to confuse? What evidence separates them?

Change one thing in the pictured case

Change: Add source B naming the same exhibit duration 20 minutes, with B declared current and authoritative.

Prediction: Duration becomes 20 minutes.

Reason: There is now an authorised source for the previously missing field.

5. Words that unlock the idea

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.





constraint



audience



Word · Meaning · My example

Word: prompt

Meaning: instruction and context given to a generative system

My example: _____

Word: constraint

Meaning: limit the output must follow

My example: _____

Word: audience

Meaning: intended reader or user

My example: _____

Word: iteration

Meaning: revising instructions after evaluating a result

My example: _____

Misconception detective

CLAIM TO REPAIR A perfect prompt can force a model to know missing facts.

Use one vocabulary word and one piece of evidence to repair it:

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word · Meaning

Word: Criterion

Meaning: A stated rule used to judge a response

Word: Test case

Meaning: A particular input with expected behaviour used to check a process

Word: Revision

Meaning: A documented change intended to address a specific problem

6. Follow a worked example

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable evidence.



CASE 'Write a story' gives little direction; a goal, audience, setting, length, required facts, and safety constraints shape a more useful output.

STEP 1 — OBSERVE goal, context, constraints, and requested format

STEP 2 — TRACE a generative model predicts an output matching the instruction

STEP 3 — CONCLUDE draft that still requires review

MODEL REASONING The improved prompt should make success observable: who it is for, what it must include, what it must avoid, and how it should be structured. Evaluate usefulness and verify factual claims separately.

DO NOT OVERCLAIM Prompt quality improves control and usefulness, not guaranteed truth.

What new evidence could change this answer?

A closer explanation

Test input · Expected behaviour · What it reveals

Test input: C plus a stated 3 pm time

Expected behaviour: Use 3 pm for Time

What it reveals: Can use supplied information

Test input: Original C with no time

Expected behaviour: Time: not stated

What it reveals: Does not fill a gap by guessing

Test input: C and D disagree on room

Expected behaviour: Flag the room conflict

What it reveals: Does not silently choose a record

For the third case, D describes the same event as C but gives Room 9, and neither record is marked as replacing the other. P1 explains missing values but says nothing about disagreement. Revise that weakness: if supplied records conflict about a requested field and no priority is given, write “conflicting information” for that field. Preserve values that do not conflict.

Call the revision P2 and keep the rest of P1 unchanged, except that it now refers to the supplied records rather than C alone. The place field should flag the conflict; it must not silently choose Room 7, choose Room 9 or combine them into Room 79. A later authoritative correction may resolve the conflict, but this input does not provide one.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Prompts Are Design Instructions

A fictional club needs a short exhibit card, without guesses.

The facts we will use

Source A: exhibit Paper Birds, location Shelf 2. No duration appears.

Required output: two labelled lines, Exhibit and Duration; missing values must say not supplied.

Prompts Are Design Instructions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State task

Produce exactly two labelled lines using source A only.



2. Find evidence

A supplies Paper Birds but no duration.



3. Follow missing rule

Exhibit: Paper Birds. Duration: not supplied.



4. Check contract

Check labels, two-line format and each factual claim.

CHANGE ONE THING

Add source B naming the same exhibit duration 20 minutes, with B declared current and authoritative.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. State task

What happens and why: Produce exactly two labelled lines using source A only.

Step: 2. Find evidence

What happens and why: A supplies Paper Birds but no duration.

Step: 3. Follow missing rule

What happens and why: Exhibit: Paper Birds. Duration: not supplied.

Step: 4. Check contract

What happens and why: Check labels, two-line format and each factual claim.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A useful prompt describes both the wanted answer and what to do when evidence is insufficient. Asking for a duration does not create one. Examples can demonstrate the format, but they must not become sources of event facts. Check content support separately from whether the answer looks tidy.

What this example does not establish

Even a complete prompt does not guarantee an AI system follows it; test returned outputs against the contract.

7. Find and repair a mistake

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case · Evidence · What it shows

Case: Write a story

Evidence: goal underspecified

What it shows: vague prompt

Case: 300-word space mystery for age 8

Evidence: audience and format clear

What it shows: better direction

Case: Use only these facts

Evidence: evidence boundary

What it shows: still verify

Case: Be accurate

Evidence: wish without checking plan

What it shows: not a guarantee

Repair two tempting claims

CLAIM A A perfect prompt can force a model to know missing facts.

Repair:

CLAIM B Adding more words always makes a prompt better.

Repair:

STRONG ANSWER FRAME I claim _____ because the evidence shows _____. The process is _____. One limit is _____.

A closer explanation

Suppose a revised prompt handles three classroom examples correctly. You can report that it passed those examples under the recorded conditions. You cannot conclude that it will always work. If outputs can vary between runs, plan repeated trials and keep every result. A single good response is not a reliability guarantee.

Use the failures to improve the specification and add revealing cases. Keep the same evaluation criteria when comparing versions; changing both the prompt and the scoring rule makes the comparison hard to interpret. A new requirement may justify a new criterion, but document that change rather than pretending the results used the same test.

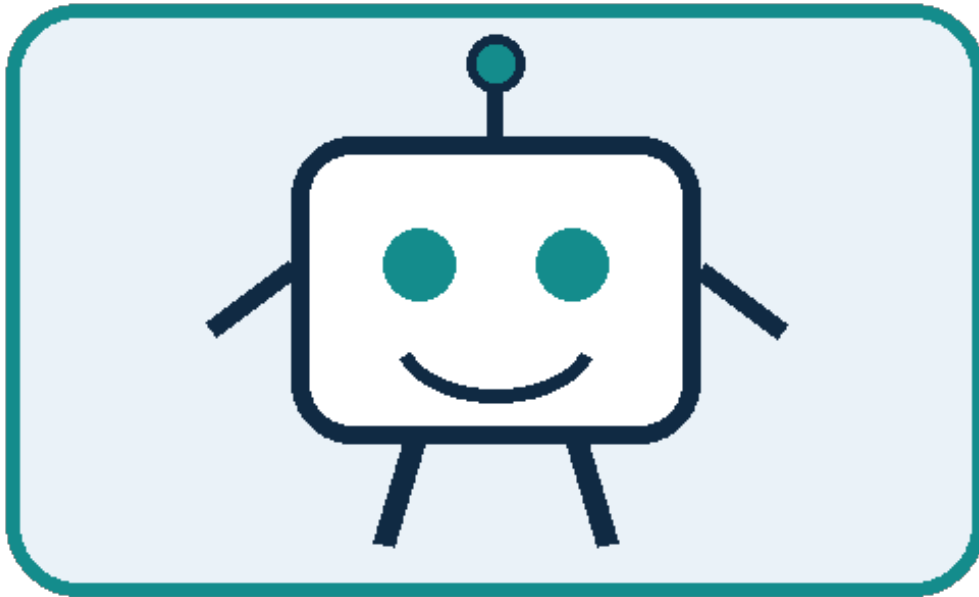
Two prompts are compared: P1 receives a complete record and P2 receives a record with a missing time. Why is that insufficient to attribute different results to the prompt change? Describe a fairer comparison

8. Practise together

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Clear goals, context, and constraints improve usefulness, not truth guarantees.

Use a fresh example to explain prompt.

Trace a new input → process → output example.

Find and repair the error in: Adding more words always makes a prompt better.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the evidence. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

A small test log can record prompt version, input case, output, each criterion's result and the reason for failure. If you use an approved model, also record the model and available settings. Compare prompt versions on the same cases under the same recorded conditions. Keep all observed outputs rather than only the best-looking attempt.

For original C, an example response has four lines: Day: Monday; Time: 3 pm; Place: Room 7;

Bring: paper. Identify the criteria it passes and fails. Explain why adding “be accurate” alone is not enough, and propose a missing-value test to rerun after a revision

P1 already tells the model not to invent missing values. A failed output can therefore be a failure to follow an adequate instruction, not necessarily a missing instruction. Inspect the actual output, retain the failure and try targeted changes if useful. Clear prompting helps communicate requirements but cannot guarantee correctness, remove the need for checks or grant permission to use restricted data.

Practise with less help: Score a response by separate criteria

Use workbook W2 for the connected example “Specifying and testing prompts”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Specifying and testing prompts. Proposed explanation: “A tidy response is a correct response”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What must happen if B instead conflicts with another equally authoritative duration record?

My explanation using a pictured step and the given facts

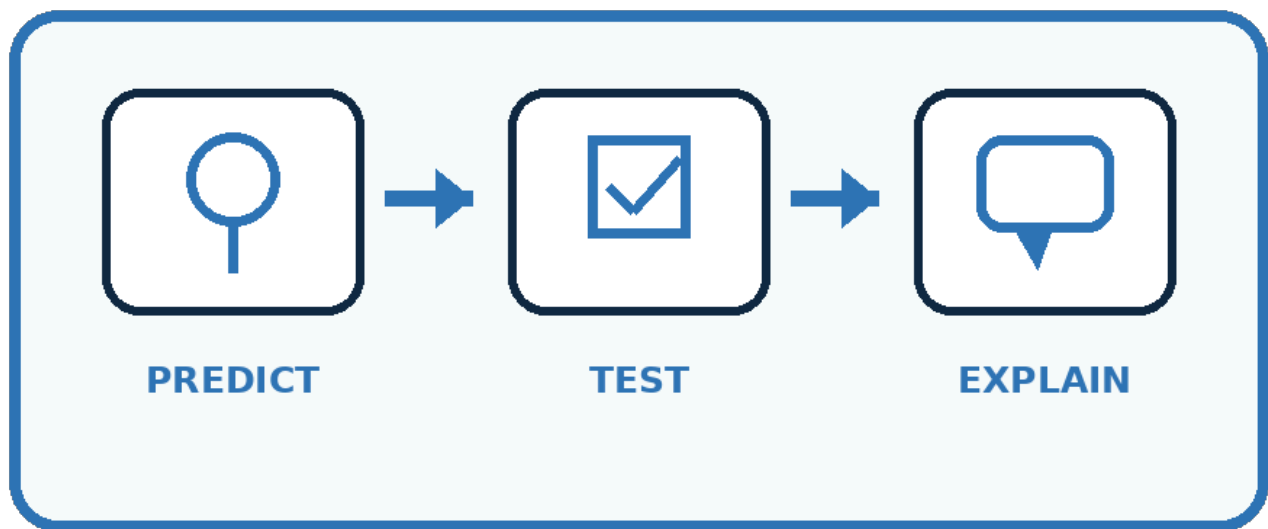
This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EXPLORER LAB

Prompt Revision Studio

Predict first, test honestly, and explain what the evidence means.



State your prediction before the result is known.

Complete the task: Revise a vague story prompt and compare both outputs.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

Case · Prediction · Observed result · What it means

Case: Normal

Prediction: _____

Observed result: _____

What it means: _____

Case: Edge

Prediction: _____

Observed result: _____

What it means: _____

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

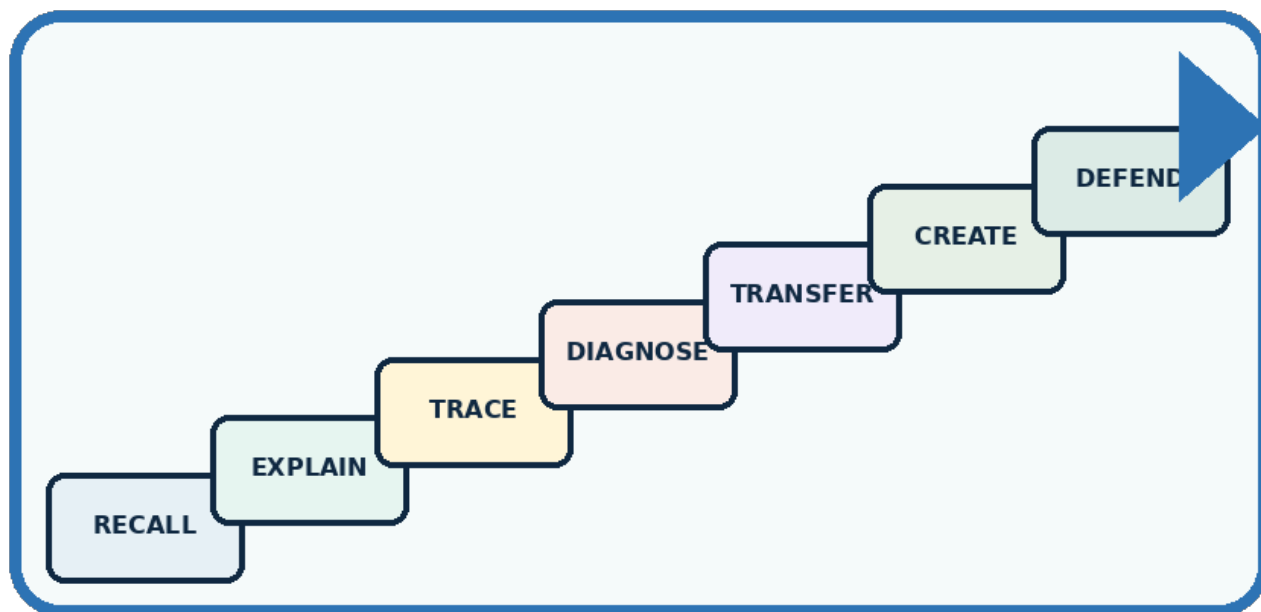
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define prompt.

EXPLAIN Clear goals, context, and constraints improve usefulness, not truth guarantees.

TRACE Show the input, process, and output.

DIAGNOSE Repair: A perfect prompt can force a model to know missing facts.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, evidence, mechanism, limit, and safe action.

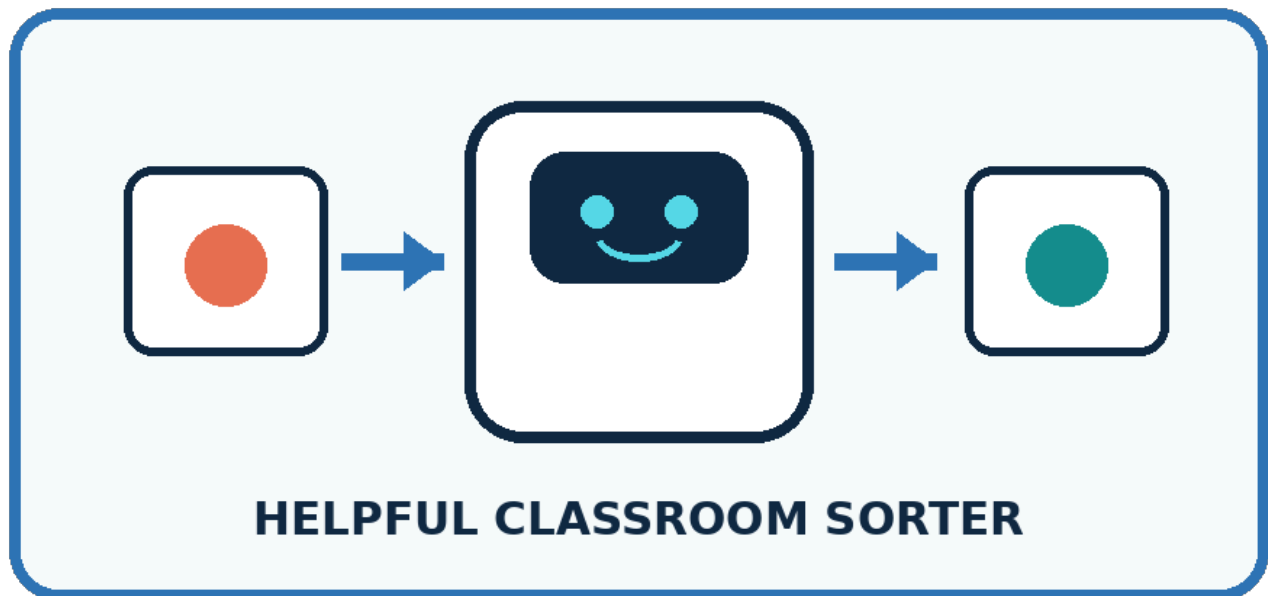
RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

11. Connect your project

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one evidence-based decision at a time.



THIS WEEK'S CHECKPOINT Write a transparent instruction for any generative explanation in the sorter and forbid invented item facts.

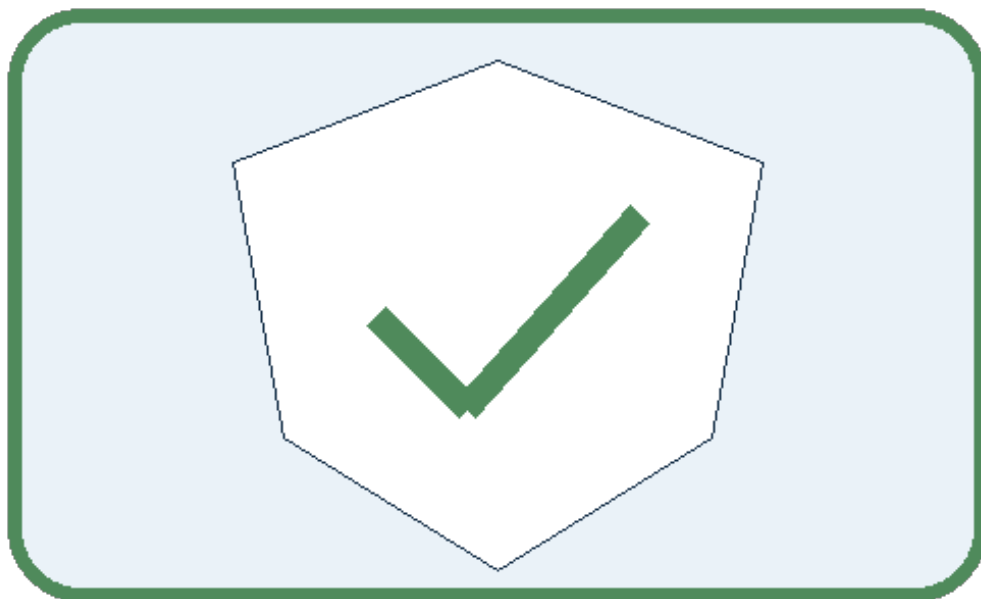
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the process?

What normal and edge cases will you test?

When should it say not sure or ask a person?



RESPONSIBLE DESIGN Prompt quality improves control and usefulness, not guaranteed truth.

My project decision:

Evidence for your project

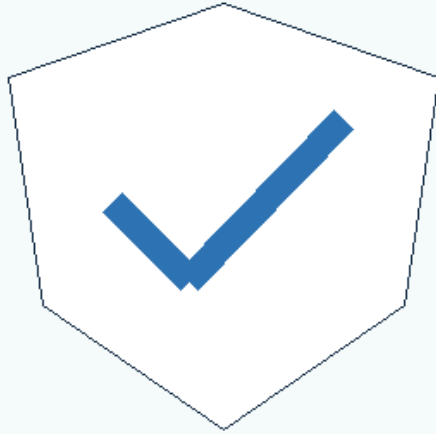
Specify sorter input, output fields, evidence limits and missing/conflict behavior before improving its wording.

12. Show what you know

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



INDEPENDENT PROOF

REQUIRED PROOF Revise a vague story prompt with goal, context, constraints, and format; compare outputs against a rubric.

- ☐ Name or draw the fresh case.
- ☐ Trace input, process, and output.
- ☐ State the conclusion and strongest evidence.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent evidence:

SELF-CHECK The improved prompt should make success observable: who it is for, what it must include, what it must avoid, and how it should be structured. Evaluate usefulness and verify factual claims separately.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

NEXT BRIDGE Carry today's evidence habit into: Verify Important Outputs.

13. Study a complete case

Week 26 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This

case extends the illustrated lesson.

The situation

The prompt says make it good; the desired output is a four-line rain poem using simple words.

The question

Write a checkable prompt.

Worked explanation

Ask for exactly four lines about rain in simple vocabulary. Check line count, topic and vocabulary separately.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Make a poem request checkable

Evidence or stage · Value or result

Evidence or stage: Length

Value or result: Exactly four lines

Evidence or stage: Topic

Value or result: Rain

Evidence or stage: Vocabulary

Value or result: Simple words

Evidence or stage: Check

Value or result: Inspect all three requirements

The prompt says make it good; the desired output is a four-line rain poem using simple words.

Write a checkable prompt.

Ask for exactly four lines about rain in simple vocabulary. Check line count, topic and vocabulary separately.

14. Try a new case independently

Independent case: Specify and test a new exhibit card

Fictional record X says an exhibit is called Shadow Pairs and is in Hall 2; no opening time is given. Write a complete prompt for a Grade 6 card with exactly three labelled lines: Title, Location, Opening. Limit facts to supplied records, define missing and conflicting values and prohibit added claims. Give two test inputs and expected outputs: original X, then X with an opening time of 10 am added.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 26 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Improve this prompt: tell me about space. Specify a reader, a focused task and an output format suitable for a young learner.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Write a transparent instruction for any generative explanation in the sorter and forbid invented item facts.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to

keep your practice evidence together.

Transfer practice from the original programme

Write a request for a three-bullet explanation of evaporation for Grade 6 using only a supplied paragraph. Create three separately checkable acceptance criteria. What should happen if the paragraph does not explain evaporation?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.